Download Titanic Dataset from the Below Link
Titanic Dataset - https://github.com/REPHEL-0101/mongodb/blob/main/titanic.csv

Questions

Pandas-Based Questions

1. What is the average age of the passengers?
2. How many passengers survived and how many did not?
3. What are the unique classes (`Pclass`) of the passengers?
4. What is the count of passengers in each class?
5. What is the gender distribution of the passengers?
6. How many passengers have missing age values?
7. What is the mean fare paid by each class (`Pclass`)?
8. List the top 5 passengers who paid the highest fare.
9. What is the most common port of embarkation (`Embarked`)?
10. What is the survival rate by gender?

Matplotlib-Based Questions

11. Plot a histogram of the `Age` distribution.
12. Create a bar chart showing the number of survivors by passenger class (`Pclass`).
13. Plot a pie chart showing the proportion of male vs female passengers.
14. Plot a boxplot of `Fare` across different classes (`Pclass`).
15. Create a line plot showing the number of passengers by age (after grouping ages into bins).